DE.AUTO

CHAPTER 6

DRIVE SYSTEMS & TORQUE

THE WORLD OF CAR MODIFICATIONS MADE SIMPLE

Cars explained in plain, friendly English

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DE.AUTO — Because knowledge is power, and cars are passion.

CAR MODIFICATIONS GUIDE
MUST-KNOW TERMS

Car modifications, explained

Chapter Six: Drive Systems and Torque

“POWER MEANS NOTHING UNTIL IT TOUCHES THE ROAD.”

— DE.AUTO

Important notice

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You must **never** try any modification, drive at high speed, or do any kind of stunt or dangerous driving on public roads. Safety and respecting traffic laws are everyone's own responsibility.

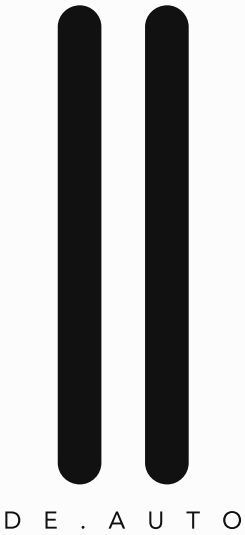
The information here is **general and approximate**, and much of it describes only the typical case. Laws, regulations and specifications differ by country and model, so always check the owner's manual and the official authorities.

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For any decision about your car, go to certified professionals and the manufacturer's manual, and only try something if it complies with official regulations and legal requirements. 🚀



DE.AUTO is more than a name. It's an identity that brings the passion for cars together with knowledge and progress.

At DE.AUTO we make fresh content for car lovers, mixing fun and know-how from the world of modifications and high performance, in a simple style that's easy to follow. Our goal is to help you understand your own car better and make smarter calls on modifications and maintenance.

We walk you through the key terms and parts of the modification world, so you can chat with the enthusiasts and follow every detail without feeling like an outsider.

Whether you're a beginner or a pro, you'll find terms, basics and fun facts here that help you get into cars in an organized, confident way.

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READY?

500 horsepower under the hood... and one wheel spinning on soft sand while the car sits still? 🤯 How does that happen?

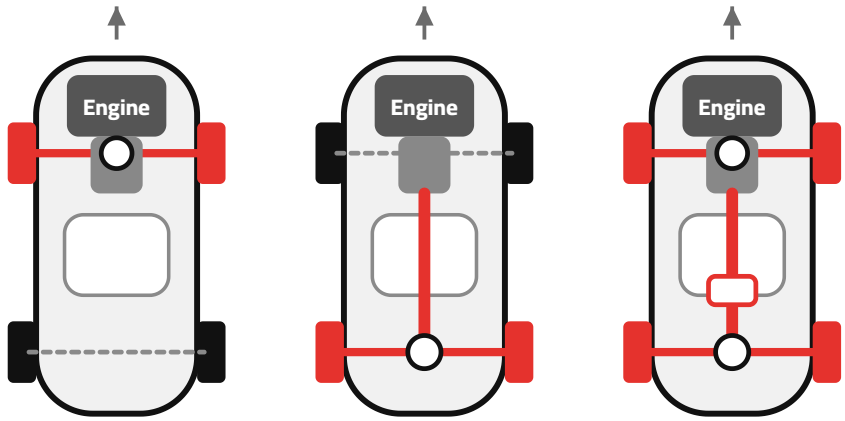
By the end of this chapter you'll know:

- ✓ Front-, rear- and four-wheel drive, and when each one wins
- ✓ Tell AWD from 4WD apart, and what “light dabl” and “heavy dabl” mean
- ✓ From engine to wheel: driveshaft, axle shaft, CV boot, final drive
- ✓ Why one wheel spins while the other stays put: the differential
- ✓ The LSD in all its types, diflok and the “heavy differential”
- ✓ Why front drive runs wide in corners and rear drive wins fans
- ✓ What gets lost between engine power and wheel power?



Where does the power go? 🛞

The engine generates the power and the gearbox arranges it, but **which wheels** receive that power? This is the “drive layout”. This one small decision changes everything: the feel of the drive, the grip, even what the engine bay looks like from underneath.



Red = the axles that receive power ▪ Front is at the top

Simplified view from above: the engine is a longitudinal front engine in these examples, and where the parts sit varies from car to car

Front-wheel drive (FWD)

Power goes to the front wheels. The most common layout in city and family cars.

Rear-wheel drive (RWD)

Power goes to the rear wheels. The classic layout in performance cars, race cars and pickups.

All-wheel drive (AWD)

Power is shared among all four wheels, usually automatically, without you doing anything.

Four-wheel drive (4WD / 4×4)

All four wheels drive, and there is usually a transfer case plus modes you choose yourself (high / low).

🧠 **The idea:** The same engine and the same gearbox in two cars, but a different drive layout, and the driving feels completely different. The tires have just **four contact patches** to work with, and you decide which of them push and which steer. 🎛️

✕ MYTH

Four-wheel drive and all-wheel drive are exactly the same thing.

✓ FACT

People use the two names interchangeably, but in engineering terms they differ: all-wheel drive **works on its own** most of the time on ordinary roads, while four-wheel drive is built more for off-road and gives you modes (including a strong low-range mode). We will separate them in a moment.

Front vs. rear: the honest truth 🥊

There is no “best” layout for everyone. Each layout wins in some places and loses in others. Here is the honest picture:

Front-wheel drive (FWD)

✅ Advantages
Lighter and cheaper to build, because everything is packed up front (engine, gearbox and differential in one unit). More room in the cabin and trunk. The engine’s weight sits over the wheels that do the driving, which gives good grip on slippery roads.

⚠️ Drawbacks
The front wheels do three jobs: they drive, they steer and they take most of the braking. If you add throttle hard, the car tends toward “understeer” in corners. And with high power you can feel a slight pull in the steering wheel on a hard launch (torque steer).

Rear-wheel drive (RWD)

✅ Advantages
The front is free to steer and the rear is free to push, so the division of labour is cleaner. Under hard acceleration weight shifts rearward, which presses the driven tires into the road. A pure driving feel, and an easier weight balance (close to 50/50).

⚠️ Drawbacks
The long driveshaft takes up room in the cabin. Under hard acceleration on a slippery surface (rain, sand, wet roads) the rear wheels can break loose more easily. And an empty, light pickup has little weight over the rear axle.

✕ MYTH
Rear-wheel drive is “dangerous” and front-wheel drive is “safe”.

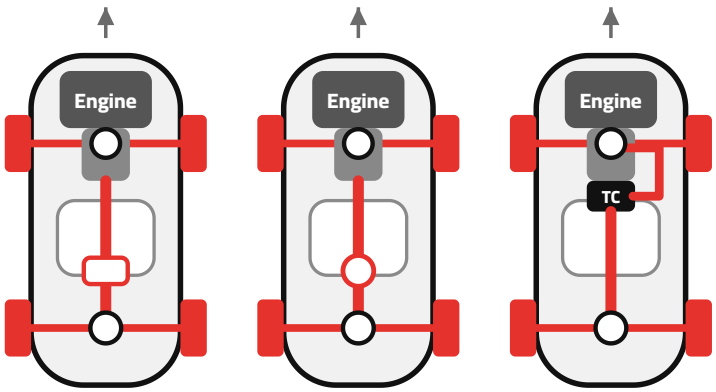
✓ FACT
The danger doesn’t come from the drive layout, it comes from **tires, speed and driving style**. Modern rear-wheel-drive cars come with stability and traction control, and good tires make a big difference. Yes, rear-wheel drive asks for more respect when it’s slippery, but it isn’t “dangerous” in itself.

60/40

Approximate weight distribution (front / rear) in many ordinary front-wheel-drive cars, because the engine and gearbox are both up front. This number is the key to how they behave in corners, and we’ll go into it on the steering page.

✅ **Before you compare:** Try the car before you judge. Sometimes the front-wheel drive you like is far smoother and more comfortable than a rear-wheel drive you don’t need in your daily life.

All-wheel drive (AWD) vs. four-wheel drive (4WD) 🧗



Three common forms; designs vary a lot between manufacturers

All-wheel drive (AWD)

A smart system for ordinary roads: it shares the power between the front and rear axles by itself. Some are permanent (relying on a **centre differential**), and some are "on demand": they mostly drive one axle, and if they sense slip they engage the other axle through a **coupling** (a clutch or a fluid).

Four-wheel drive (4WD)

Built for rough ground and heavy loads. It has a **transfer case** behind the gearbox that splits the power between two shafts, and it often includes a **slow, strong** low-range mode for tough going.

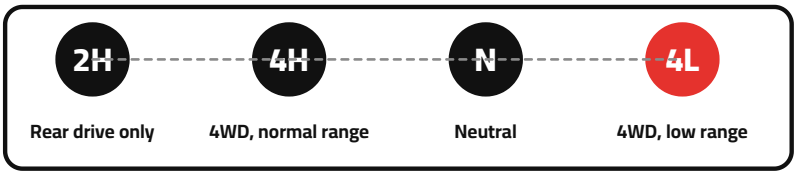
Part-time vs. permanent 🔧

Part-time

Mostly drives the rear wheels, and when you need it you switch on four-wheel drive with a button or lever. There is no centre differential, so both axles are locked together, and that means: **not suitable for dry tarmac** because the wheels have to turn at different speeds in corners.

Full-time

All four wheels work all the time through a centre differential that allows a difference in speed. Suitable for tarmac and off-road, and usually able to lock the centre differential when needed.



Example transfer-case modes; names and sequence vary between cars

4L In many four-wheel-drive vehicles the low range **"4L"** gives a big multiplication of force at the wheels (it varies from car to car by design), in exchange for very slow speed. It is for crawling over hard obstacles, heavy sand and towing, not for the road.

Is all-wheel drive always faster? 🏁

Quick answer: **no**. All-wheel drive is clearly excellent at one thing: **launching** when grip is low (rain, sand, a slippery curb), because it spreads the drive over four contact patches instead of two. But:

More weight: extra axles, shafts and parts, so the car is usually heavier.

More loss: more parts are turning, so a bigger share of the power is lost in transmission (we'll cover that shortly).

It doesn't improve braking: braking is done by the tire and the brakes, not by the drive layout.

It doesn't raise the cornering limit: in a fast corner the tires decide, and some AWD cars add safety on launch but remain heavy at the front.

Higher maintenance and cost: extra oils (differentials and transfer case), and more potential failures.

✕ MYTH	✓ FACT
All-wheel drive stops you faster and keeps you planted in corners.	All-wheel drive helps you **launch** better. Braking and stability in corners are governed by the tire, the suspension and the weight. And if your extra confidence makes you go faster, physics will remind you. Safety comes from good tires and a sensible speed.

> 💡 **Did you know?** In permanent all-wheel drive you have **three differentials**: front, rear and centre. Each one solves a different turning problem. 🧩

"Light dabl" and "heavy dabl" in garage talk 🗣️

If you hear "hit the dabl" or "heavy dabl" on trips and in the desert, don't be fooled: it isn't a new differential.

Dabl ("the double") (4WD)
In the Gulf this usually means **switching on four-wheel drive** (the button or lever that sends power to the front), not the part called the differential.

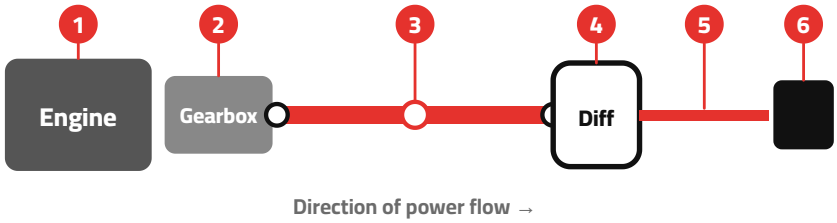
Light dabl (4H — Four High)
Four-wheel drive in normal gearing, for medium sand, dirt tracks and rain.

Heavy dabl (4L — Four Low)
Four-wheel drive in a **slow, powerful** gear: for crawling through deep sand, mud, rocks, hard climbs and towing.

Diflok (Diff Lock)
Short for "differential lock" = **locking the differential**: it makes the two wheels on the same axle turn at the same speed. We explain it fully in a moment.

✅ **A note on language:** The names vary between workshops, regions and models, so if in doubt ask, "Do you mean four-wheel drive (4 High / 4 Low) or a differential lock?" and check your car's manual.

From gearbox to wheel: the parts that carry the power ⚙️



Simplified rear-wheel-drive layout: power flows from left to right

1) Engine
Generates the power.

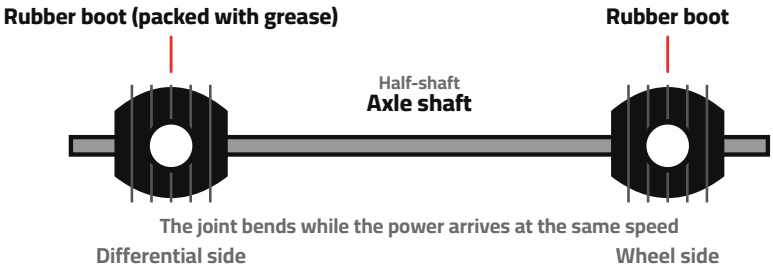
2) Gearbox
Chooses the gear.

3) Driveshaft (Propeller shaft)
A long shaft that carries power to the rear. It has flexible joints (universal joints) or CV joints. If its balance is off, you feel vibration with speed.

4) Final drive (Differential)
Has two gears (pinion and ring gear) that change the direction of the power by 90°, reduce the rotation speed and multiply the torque, and it houses the differential itself.

5) Axle shaft
Carries the power from the differential to each wheel.

6) Wheel
Turns rotation into thrust on the ground.



An axle shaft with two CV joints (illustration)

The CV joint (Constant-Velocity Joint) is what allows the wheel to steer left and right and move up and down over bumps, all while the power reaches it smoothly. Its protection is the **boot**, the rubber cover filled with grease.

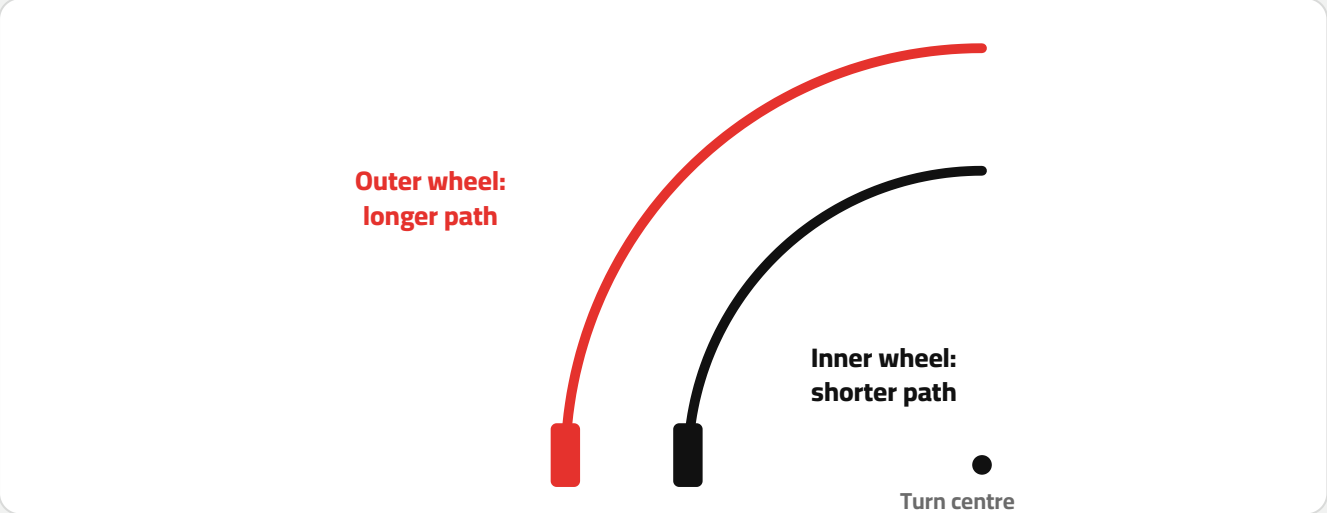
❌ **A torn boot = grease flies out = the joint wears out.** If you hear **repeated clicking** when turning slowly, or you see grease sprayed inside the rim, have it checked early. The boot is cheap, and the joint costs much more if it goes.

🎯 **Challenge 1** **What is the classic sign of a worn CV joint?**

A) Squealing when braking | B) Repeated clicking in sharp turns | C) Vibration in the pedal

The differential: the smartest part you never notice 🧠

When you go around a corner, the outside wheel travels a longer distance than the inside one in the same time. So it has to spin **faster**. If both were joined by one solid shaft, one of them would have to slip or "dig" into the ground. The differential solves this problem.



Same time, two different distances, which means two different rotation speeds

9.4 M

For example: if the distance between the two wheels (track width) is about 1.5 metres, then in one full circle the outer wheel travels about 9.4 metres more than the inner one (2 × π × 1.5), however big the circle is. This is an approximate example. Imagine that difference with no differential!

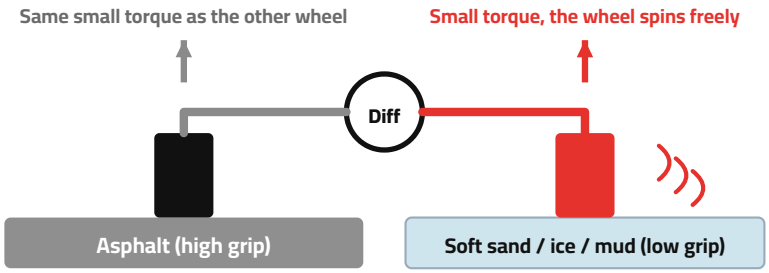
How does it work, in principle?

1. Power arrives from the driveshaft at a small gear (**pinion**) that drives a big gear (**ring gear**).
2. The ring gear turns a **carrier** that holds small gears called "spider gears".
3. The spider gears mesh with two side gears, each one connected to an axle shaft.
4. On a straight road they all turn as one piece. In a corner, the spider gears start spinning on their own axis and let one wheel speed up while the other slows down.

> 🔑 **The key idea:** The open differential splits the **torque** (the turning force) equally between the two wheels, and splits the **speed** in whatever way the corner needs. This difference between torque and speed is the key to understanding everything that follows.

The weak spot of the open differential 😬

The standard differential is called an **open differential**. It is clever in corners, but it has a famous weak spot: if one wheel is on a slippery surface, it gives both wheels **the same torque**, and the torque a slippery wheel can hold is small. The result?



An open differential gives both wheels the same torque, and the torque is limited by whichever wheel has less grip

The slippery wheel "lets go" and spins fast, the wheel on excellent ground hardly works, and the car doesn't move. This is exactly the story from the chapter opener: five hundred horsepower, and one wheel spinning uselessly. 🎯

✕ MYTH
If I have lots of power, the slippery wheel will pull the car along.

✓ FACT
Usually not. The amount of torque reaching **both sides** is limited by the lower grip. The extra power doesn't turn into thrust, it just spins the slippery wheel faster.

So what are the solutions?

1) Electronic traction control (Electronic LSD)
The car senses the wheel that's slipping and brakes it automatically, so torque moves to the other wheel. It imitates an LSD using the brakes. It works with no extra parts, but it heats up the brakes if it goes on for long.

2) Limited-slip differential (LSD)
A mechanical part that limits the speed difference between the two wheels and passes torque to the wheel that's gripping.

3) Differential lock (Locker)
Ties the two wheels together rigidly, so they turn at the same speed whatever the ground. For off-road.

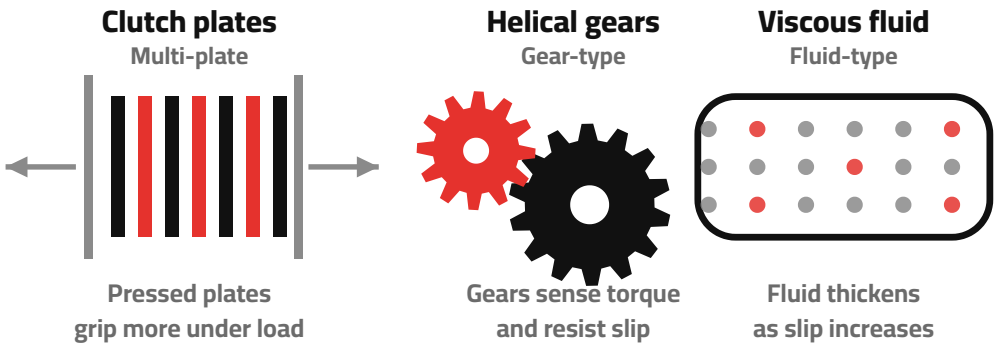
4) Better tires
Many "wheelspin" problems are solved by a tire that suits the surface and the right air pressure before any part is changed. (Chapter 10).

🎯 Challenge 2 **One wheel on asphalt and one on soft sand, and the car has an open differential. What usually happens?**

A) The gripping wheel takes all the torque
B) The slippery wheel spins and the car barely moves
C) Nothing happens

• Limited-slip differential (LSD) 🔒

The idea: **a differential that behaves open in corners and locks up when it senses slip**. So it's the best of both worlds… to a degree. There are three common types (a general description; each manufacturer has its own design):



Simplified concept sketches; real designs vary a lot between manufacturers

Type	Idea	Pros and cons
Clutch-type	Friction plates pressed by springs and load	Sharp response and adjustable, but it wears, can make noise or feel rough, and usually needs a specific oil.
Helical	Gears that sense torque	Smooth and durable with no plates to wear, but it needs grip on both wheels. If one wheel is lifted in the air, it won't transfer anything.
Viscous	A fluid that stiffens with speed difference	Quiet and easy, but slow to respond, and it is affected by heat over time.

> 🔑 **Remember:** An **LSD** is not a "full lock". It reduces wheelspin and helps you power out of a corner, without becoming a solid lock on tarmac. 🎯

What do they improve?

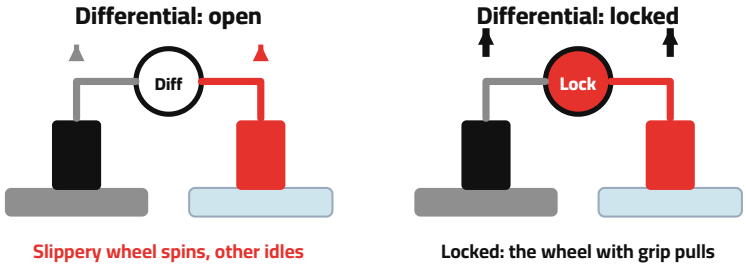
Corner exit: helps the inside wheel not spin when you get on the throttle after the apex.

Stability under power: spreads torque better on powerful cars, especially on a track.

Not meant for fast driving on open roads: its behaviour when it's slippery differs from what you're used to, so respect the system, the tires and the conditions.

• Diflok: fully locking the differential 🔐

Diflok is short for “diff lock”. The name is common in the Gulf and simply means **locking the differential**: instead of distributing speed, the differential ties the two wheels together rigidly. Both wheels turn at exactly the same speed, so if one is on gripping ground, it pulls the car.



The difference between an open and a locked differential on ground that is half grip and half slip

Mechanical / automatic (Automatic Locker)
Locks itself when it senses a speed difference, and sometimes you hear “clicking” in corners. Excellent for hard work, but the feel is rough.

Manual / electronic (Selectable / Electronic Locker)
You press a button (or the computer engages it) and it locks. You release it when the off-road section is over. Some systems release it automatically above a certain speed.

Axle lock
Can be on the rear axle, the front, or both. A front lock improves steering off-road but needs care.

Centre differential lock
In permanent four-wheel drive it ties the front axle to the rear instead of allowing a speed difference between them.

❌ **Leaving the diflok locked on tarmac.** A locked differential prevents speed differences in corners, so the shafts, gears and tires get stressed, and the car “hops” and fights the steering. Use it only on clearly off-road ground at slow speeds, and release it when you return to tarmac.

• “Heavy” differential (heavy-duty) 🏋

There are two meanings you need to tell apart:

“Heavy dabl” (desert talk) (4L)
Usually means the low range **4L** in four-wheel drive, not a strengthened differential. We covered it above.

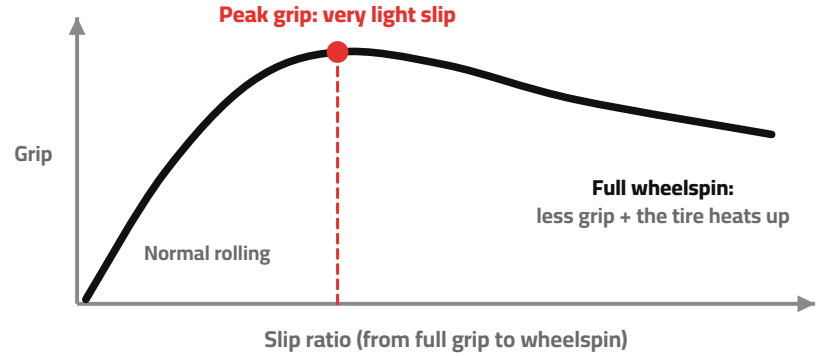
Strengthened differential (engineering) (Heavy-Duty Differential)
A bigger, stronger differential: a bigger gear, a stronger case, stronger shafts and better cooling. Essential for heavy towing, constant loads, high torque and sustained off-road work.

🎯 Challenge 3 **“Heavy dabl” in desert-trip talk usually means?**

A) A bigger rear differential
B) Four-wheel drive in the slow, strong low range (4L)
C) Locking the front wheel only

Grip and wheelspin 🧊

The tire sets how much power it can pass to the road. Beyond that limit you get “wheelspin”: the rotation turns into heat and smoke instead of pushing you forward. And the surprise? **Peak grip comes just before full wheelspin, not after.**



Illustrative qualitative curve (not real numbers): peak grip comes before full wheelspin, not after

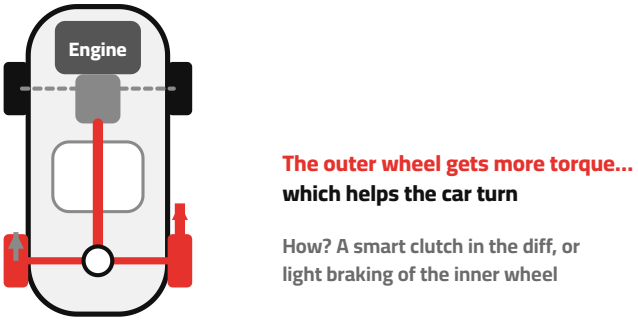
Wheelspin on launch: the driven wheels spin faster than the car. Traction control cuts the throttle automatically.

Why is a gentle launch sometimes quicker? because the tire stays close to the point of peak grip.

The ground changes the rule: in sand and wet there is less grip, so a light touch matters even more.

Torque vectoring 🎛️

A smart, modern idea: instead of splitting the torque equally, we give **the best-placed wheel** more torque depending on its situation. In a corner the outer wheel carries more load and can take more, and giving it more push helps the car “turn in” more smoothly.

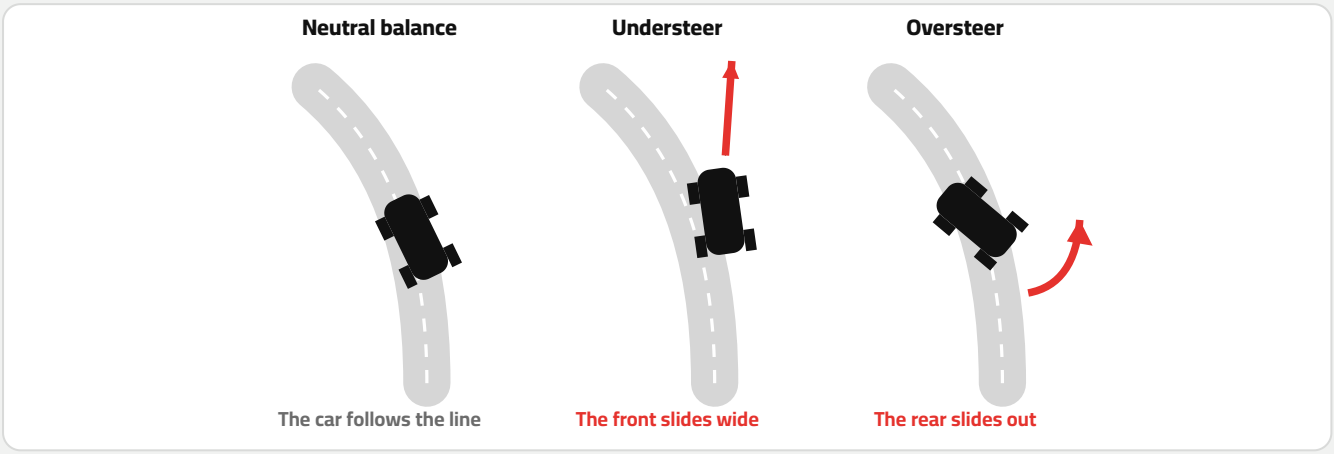


The idea of torque vectoring: more torque to the outer wheel in a corner

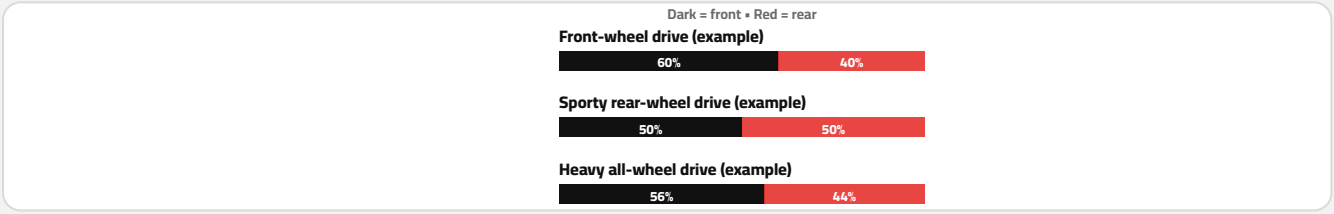
> 💡 **Did you know?** Some systems do it with an active differential containing clutches the computer controls, and some imitate it by lightly braking the inner wheel. Same goal, two methods.

Weight and steering: why does every car have a character? 🌀

The four tires have a "grip budget". The more you ask of them (driving, steering, braking), the closer they get to the limit. How the weight is distributed and the drive layout decide which tire reaches the limit first.



A left-hand corner seen from above; the red arrow shows where the car actually goes



Illustrative numbers to give the idea; they vary from car to car

Understeer

The car keeps going straight more than you steered, because **the front tires** have reached their limit. The fix: ease off the throttle and let the tires grip again.

Oversteer

The rear steps out and turns the car more, because **the rear tires** have reached their limit. It takes calm and a smooth correction, and stability systems step in early.

Why does front-wheel drive lean toward understeer?

The engine's weight is up front, and the front tires drive, steer and brake all at once. If you ask all three of them hard, the "budget" runs out fast and they slide wide. It's a predictable character and easy to correct, which is why manufacturers choose it so often.

And why is rear-wheel drive "sporty"?

The front is free to steer, the rear is free to drive, and the weight is closer to balanced. So you feel more precision and cleaner control with the throttle. That is why track fans love it (on tracks and organised events only).

🎯 Challenge 4 **Why do front-wheel-drive cars tend toward "understeer"?**

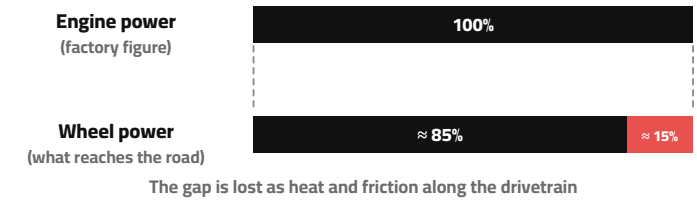
A) The front tires drive, steer and brake at the same time

B) The rear is lighter than it should be

C) The differential is open

Drivetrain loss: why do the numbers differ? 📉

The figure the manufacturer publishes is the power of the **engine** (at the flywheel). But what reaches the road is less, because the power passes through the gearbox, shafts, differential and wheels, and each part steals a bit through friction and heat. It's called **drivetrain loss**.



Illustrative example with a roughly common percentage; the real figure varies between cars

≈ 15% A common approximate figure for power lost between the engine and the wheels (usually somewhere between 10% and 20%), varying with the type of gearbox and the drive layout. All-wheel drive usually loses more than front or rear drive, because more parts are turning.

Engine power (Crank hp)
Measured at the engine itself.

Wheel power (WHP)
Measured on a performance test machine (the dyno), and it's lower.

⚠️ **Careful:** Don't compare a factory (engine) figure with a dyno (wheel) figure as if they were the same thing. This is a very common mistake in discussions. In **Chapter 13: Racing and Performance Measurement** we'll explain how the dyno works and how to read its numbers accurately.

🎯 Challenge 5 **A car's engine power is 300 hp and its drivetrain loss is about 15%. Roughly how much reaches the wheels?**

A) 285 hp B) 255 hp C) 315 hp

Drivetrain "mods"? (just the idea) 🧰

Tires first: the biggest effect on launch and grip comes from the right tire and the right air pressure, before any part.

Limited-slip differential: used to improve corner exit and high power, and it needs specialist fitting and the right oil.

Differential and gearbox oils: stick to the type specified by the manufacturer, especially with an LSD.

Any change to the drivetrain can affect the warranty, insurance and inspection, so ask a certified specialist and check the local regulations. No part gives you a guaranteed figure.

Common mistakes and myths 🚫

✕ MYTH	✓ FACT
Four-wheel drive means I'll never get stuck in sand.	Four-wheel drive helps, but getting stuck comes down to the tires (air pressure), driving style and the ground. A reckless or inexperienced driver gets stuck even in the strongest car.
✕ MYTH	✓ FACT
I engage the dabl (4WD) on dry tarmac to be more stable.	In part-time four-wheel drive this strains the shafts and wears parts out, because the wheels need to turn at different speeds. Use it for sand, dirt and slippery surfaces, according to your car's manual.

❌ **Fitting tires of different sizes or different wear on an all-wheel-drive car.** A difference in tire diameter makes the shafts and the coupling work under constant strain. Replace all the tires, or follow the manufacturer's instructions.

❌ **Neglecting the differential, gearbox and transfer case oil.** Oil tires out with heat and dust, and each part has its own specification. The wrong oil in an LSD causes noise and vibration.

❌ **Ignoring a clicking noise from the joints or driveshaft vibration.** Early repair = much cheaper.

❌ **Going off-road without knowledge.** Designated areas and organised events are best, with companions, recovery gear and knowledge of the terrain.

Garage talk vs. the proper name 🗣️

Gulf garage talk	Proper name	What it means
Dabl	Engaging four-wheel drive (4WD)	A lever or button that sends power to the front
Dabl khafeef / thaqeel (light / heavy)	4H / 4L	4WD in normal gearing / in slow, strong low gear
Diflok / the lock	Diff Lock	Differential lock: both wheels at the same speed
Difrens / Difrishin	Differential	Shares the power and allows a speed difference
Heavy differential	Heavy-Duty Differential	A reinforced differential for loads and high torque
Jilda	Joint cover (CV Boot)	A rubber cover that protects the joint and its grease

Wrap-up 📌

Now you know that power doesn't end at the gearbox: it has to pass through the driveshaft, the differential and the axle shaft before it touches the ground. You can tell front, rear, all-wheel and four-wheel drive apart, you know exactly what "heavy dabl" and "diflok" mean, why the open differential lets one wheel spin, and how the LSD and the lock solve it. And you understand weight, steering and drivetrain loss. 🚀

🎯 Challenge answers:

1) Repeated clicking in sharp turns.

2) The slippery wheel spins and the car barely moves.

3) Four-wheel drive in the slow, strong low range (4L).

4) The front tires drive, steer and brake at the same time.

5) About 255 hp (300 × 0.85).

➡️ SOON In **Chapter 7: Turbochargers and Superchargers**, we'll see how to push more air into the engine and raise its power, and what happens to pressure and heat. Ready?